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DEPARTMENT OF ENVIRONMENTAL HEALTH
FACTORS AFFECTING HOUSEHOLD LATRINE
UTILIZATION IN MIZAN AMAN TOWN, SOUTHWEST
ETHIOPIA

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Abstract

Background: Although latrine coverage has expanded in many parts of Ethiopia, the presence of a latrine does not guarantee its consistent use, and non-utilization continues to expose communities to preventable faeco-oral diseases. Little empirical

evidence exists on household latrine utilization in Mizan Aman Town, Southwest Ethiopia.

Objective: This study will assess the magnitude of household latrine utilization and identify the socio-demographic, socio-economic, knowledge- and attitude-related, environmental and cultural factors affecting it in Mizan Aman Town.

Methods: A community-based cross-sectional study will be conducted among 423 latrine-owning households selected using a multistage sampling technique. Data will be collected through interviewer-administered, pretested structured questionnaires supplemented by an observation checklist, entered into EpiData and analysed using SPSS version 26. Descriptive statistics and binary logistic regression will be applied, and variables with a p-value of less than 0.05 will be declared statistically significant.

Significance: The findings will provide evidence-based information for the Town Health Office, WASH partners and policy-makers to design targeted interventions that promote sustained latrine utilization.

Keywords: *Household latrine; latrine utilization; sanitation; WASH; Mizan Aman; Ethiopia*

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List of Abbreviations and Acronyms

Abbreviation	Full Meaning
AOR	Adjusted Odds Ratio
CI	Confidence Interval
CLTSH	Community-Led Total Sanitation and Hygiene
COR	Crude Odds Ratio
CSA	Central Statistical Agency
EPHI	Ethiopian Public Health Institute
ETB	Ethiopian Birr
HEP	Health Extension Program
IRB	Institutional Review Board
JMP	Joint Monitoring Programme
OWNP	One WASH National Program
SDG	Sustainable Development Goal
SPSS	Statistical Package for the Social Sciences
SSA	Sub-Saharan Africa
UNICEF	United Nations Children's Fund
WASH	Water, Sanitation and Hygiene
WHO	World Health Organization

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

Sanitation, which refers to the safe management of human excreta, solid waste and wastewater, is universally recognized as one of the most fundamental determinants of public health and human dignity (WHO, 2018). At the global level, despite remarkable progress since the adoption of the Millennium Development Goals, sanitation coverage still lags far behind expectations. According to the WHO/UNICEF Joint Monitoring Programme, an estimated 3.5 billion people worldwide still lacked safely managed sanitation services in 2022, approximately 419 million people continued to practise open defecation, and about 2 billion people lacked basic sanitation facilities such as private improved latrines (UNICEF & WHO, 2023). In response to this challenge, the international community committed itself, under Sustainable Development Goal (SDG) target 6.2, to achieve adequate and equitable sanitation and hygiene for all and to end open defecation by 2030, with special attention to the needs of women, girls and vulnerable groups (United Nations, 2015).

The public-health importance of sanitation arises from the central role that faecal contamination plays in the transmission of infectious diseases. Inadequate disposal of human excreta facilitates the transmission of diarrhoeal diseases, intestinal helminths, typhoid fever, cholera, hepatitis A, trachoma and poliomyelitis (Mara et al., 2010; Bartram & Cairncross, 2010). It has been estimated that inadequate water, sanitation and hygiene (WASH) was responsible for approximately 842,000 diarrhoeal disease deaths annually in low- and middle-income countries, of which more than 360,000 occurred among children under five years of age (Prüss-Ustün et al., 2014). Systematic reviews have consistently demonstrated that improved sanitation substantially reduces the risk of diarrhoeal morbidity (Fewtrell et al., 2005; Wolf et al., 2014). Importantly, the World Health Organization emphasizes that these benefits can only be realized when sanitary facilities are not only available but are also consistently and correctly used by all household members (WHO, 2018).

The economic dimension of the sanitation challenge reinforces its public-health urgency. Households affected by preventable faeco-oral diseases incur direct costs for consultation, medicines and transport, and indirect costs through lost working days and

reduced agricultural productivity. At the national scale, these losses aggregate into significant drains on already constrained health budgets, which is why international frameworks now treat sanitation not merely as a social service but as an investment with measurable economic returns (Bartram & Cairncross, 2010; WHO, 2018). Maximizing those returns requires that existing and newly built facilities be used consistently, a condition that cannot be assumed and must therefore be measured and understood.

The situation in sub-Saharan Africa (SSA) is particularly concerning. The region carries a disproportionate share of the global sanitation burden: roughly seven out of every ten people who practise open defecation live in SSA or Southern Asia, and regional progress towards safely managed sanitation has been slower than population growth (UNICEF & WHO, 2023). Rapid urbanization has further compounded the problem, as growing informal settlements in secondary towns are frequently served by unimproved shared pits and unsafely managed onsite systems. Consequently, faeco-oral diseases remain leading causes of outpatient visits, hospital admissions and child mortality across much of the continent (Bartram & Cairncross, 2010).

Ethiopia is one of the countries that has made concerted efforts to expand sanitation coverage through large-scale programmes. The Health Extension Program (HEP), launched in 2003, deployed trained Health Extension Workers to deliver preventive health packages, including hygiene and environmental health, at the community level (Federal Ministry of Health, 2015). The country subsequently adopted Community-Led Total Sanitation and Hygiene (CLTSH) approaches and the multi-sectoral One WASH National Program (OWNP) to accelerate the elimination of open defecation and increase household latrine coverage (Kar & Chambers, 2008; Ministry of Water, Irrigation and Energy, 2013). These efforts have contributed to a steady expansion of self-reported latrine ownership. Nevertheless, according to the Ethiopian Demographic and Health Survey, only about 35 percent of households nationally used improved toilet facilities that were not shared with other households, with markedly lower coverage in rural areas than in urban centres (Central Statistical Agency & ICF, 2016). The Ethiopia Mini Demographic and Health Survey confirmed that sanitation coverage and practice continue to vary widely across regions and residence types (Ethiopian Public Health Institute & ICF, 2021).

Southwest Ethiopia is characterized by a humid tropical climate, high annual rainfall and predominantly agrarian livelihoods based on coffee and spice production.

Mizan Aman Town, the administrative centre of Bench Sheko Zone in the Southwest Ethiopia Peoples' Regional State, lies approximately 560 kilometres southwest of Addis Ababa. Like many fast-growing secondary towns in the region, Mizan Aman is experiencing rapid population growth driven by commercial activity, civil service expansion and rural-urban migration [VERIFY LOCALLY: current population size, number of kebeles and number of households from the Town Administration]. Most residents rely on onsite sanitation facilities, principally simple and ventilated improved pit latrines, while a limited sewerage network serves only a small proportion of the town [VERIFY LOCALLY: latrine coverage and sewerage statistics from the Town Health Office]. High rainfall and a shallow water table further increase the risk of groundwater contamination and latrine collapse where facilities are poorly constructed or poorly maintained.

It is increasingly recognized, both globally and in Ethiopia, that increasing latrine coverage does not automatically translate into improved health outcomes. Studies conducted in different parts of the country have repeatedly shown that a considerable proportion of households that possess latrines do not use them consistently, either because the facilities are incomplete, dirty, unsafe or socially unacceptable, or because long-standing open-defecation habits persist (Anteneh & Kumie, 2010; Yimam et al., 2014; Koyra et al., 2017). Understanding why households fail to utilize the facilities they already own is therefore essential for converting coverage gains into genuine public-health benefits. It is against this background that the present study will investigate the factors affecting household latrine utilization in Mizan Aman Town.

1.2 Statement of the Problem

Global progress towards sanitation targets, while real, remains uneven and insufficient. Between 2015 and 2022 the proportion of the world population using safely managed sanitation rose only modestly, and at current rates of change most developing regions will miss the 2030 target by wide margins (UNICEF & WHO, 2023). Monitoring agencies have therefore emphasized that accelerating progress requires understanding not only where facilities are absent, but also where existing facilities are unused — an information need that national surveys alone cannot satisfy at the level of individual towns.

A persistent paradox in sanitation programming is the gap between latrine ownership and latrine use. Households may construct latrines to satisfy programme targets or social expectations, yet continue to defecate in the open environment, in nearby bushland or along riverbanks. Evidence from Ethiopian studies indicates that utilization rates among latrine-owning households range widely, and that facilities are frequently found dirty, damaged, without doors or roofs, or otherwise unfit for regular use (Anteneh & Kumie, 2010; Koyra et al., 2017). Such partial utilization undermines the epidemiological rationale of sanitation investments, because even a minority of households practising open defecation can sustain the environmental transmission of enteric pathogens within a community.

The consequences of poor latrine utilization are severe and multidimensional. Faeco-oral diseases such as acute watery diarrhoea, typhoid fever, intestinal parasites and helminthic infections remain among the leading reasons for outpatient attendance in many Ethiopian towns, imposing substantial direct and indirect costs on households and overstressing primary health-care facilities (Prüss-Ustün et al., 2014; Wolf et al., 2014). Children, whose play and defecation behaviours differ from those of adults, bear a disproportionate share of this burden, and repeated episodes of diarrhoea contribute to malnutrition, stunting and impaired cognitive development. Women and girls additionally suffer safety, privacy and dignity-related harms when latrines are unavailable or unusable (UNICEF & WHO, 2023).

Although national attention has historically focused on rural open defecation, secondary towns such as Mizan Aman present distinct and growing challenges. Rapid in-migration, dense settlement in peripheral kebeles, high land pressure, limited municipal finance and weak enforcement of housing and sanitation by-laws combine to produce sanitation conditions that resemble those of rural areas while carrying higher epidemic risks because of population density. In such settings, latrines may be shared, poorly maintained, prone to flooding during the rainy season, or abandoned altogether once they fill up [VERIFY LOCALLY: town sanitation profile from the Town Health Office]. Whether, and how intensively, households in Mizan Aman actually use the latrines they own is currently unknown.

Furthermore, the majority of published latrine-utilization studies in Ethiopia have been conducted in rural districts of Amhara, Oromia, SNNPR and Tigray regions, focusing on model villages or CLTSH-triggered communities (Anteneh & Kumie, 2010;

Yimam et al., 2014; Koyra et al., 2017). Comparative evidence from urban and peri-urban communities in Southwest Ethiopia is scarce, and no published study could be identified that quantifies the magnitude and determinants of household latrine utilization in Mizan Aman Town. This evidence gap constrains the ability of the Town Health Office, the Town Water and Sewerage Service Enterprise and their partners to design targeted, behaviour-centred sanitation interventions and to monitor progress towards SDG target 6.2.

A further layer of the problem concerns urban sanitation governance itself. Secondary towns frequently fall between policy categories: they are too dense and commercially oriented for rural CLTSH approaches to fit comfortably, yet too small and resource-constrained to operate full municipal sewerage systems. In such towns, responsibility is shared between households, landlords, kebele administrations and utility enterprises, and no single actor possesses complete information about whether existing facilities are actually serving their intended purpose. Without household-level utilization data, by-law enforcement and hygiene promotion risk being misdirected, and scarce public resources may continue financing construction programmes whose facilities quietly fall out of use.

Therefore, this study will assess the magnitude of household latrine utilization and identify the socio-demographic, socio-economic, knowledge- and attitude-related, environmental and cultural factors affecting it in Mizan Aman Town. The findings will help to close the local evidence gap and provide actionable information for promoting consistent and correct latrine use.

1.3 Significance of the Study

The findings of this study will have practical significance for sanitation governance in Mizan Aman Town. By quantifying the magnitude of latrine utilization and identifying the specific barriers and facilitators operating at household level, the study will enable the Town Health Office and Health Extension Workers to move beyond coverage-based reporting towards utilization-based programming. Interventions such as latrine upgrading campaigns, hygiene promotion sessions, behaviour-change communication and enforcement of sanitation by-laws can then be targeted at the household segments and facility deficiencies identified as most influential.

At the regional and national levels, the study will generate comparable evidence from an understudied urban setting in Southwest Ethiopia, thereby enriching the national knowledge base on sanitation behaviour. The results will support regional health bureaus, the Bench Sheko Zone Health Department and WASH partners in reviewing the adequacy of existing CLTSH and OBNP strategies when applied in town contexts, and in allocating resources to interventions with the greatest potential to increase consistent utilization.

For the scientific community, the study will contribute context-specific data on an important but neglected dimension of urban sanitation, and will serve as a baseline against which future intervention studies and routine monitoring exercises can be evaluated. Methodologically, the instruments developed and validated in this study, including the structured questionnaire and observation checklist presented in the annexes, can be adapted by subsequent researchers working in comparable settings.

Finally, the study will benefit the participating community itself. The interview process includes brief hygiene education messages delivered by trained enumerators, and the aggregated findings will be disseminated to community representatives and town stakeholders, thereby raising awareness of the importance of consistent latrine use and stimulating community-led action towards a cleaner and healthier living environment. Because women and girls carry the heaviest burden when latrines remain unused, particular attention will be paid in analysis and interpretation to factors of gendered relevance, so that the resulting recommendations contribute to dignity, safety and equity as well as to health.

CHAPTER TWO: LITERATURE REVIEW

This chapter reviews theoretical and empirical literature relevant to household sanitation and latrine utilization. It begins by clarifying the concept of household sanitation, then defines household latrine utilization and its public-health importance. It subsequently examines the factors that influence latrine utilization as reported in the international and Ethiopian literature, summarizes findings of previous studies, identifies the research gap, reviews the theoretical perspectives underpinning the study, and presents the conceptual framework that will guide the investigation.

2.1 Concept of Household Sanitation

Sanitation is broadly defined as access to facilities and services that safely dispose of human urine and faeces, thereby interrupting the transmission of excreta-related infections (WHO, 2018). In the monitoring framework of the WHO/UNICEF Joint Monitoring Programme, sanitation services are classified along an incremental ladder: safely managed services, basic services, limited services, unimproved facilities and open defecation. A household is considered to have basic sanitation when it uses an improved facility that is not shared with other households, while safely managed sanitation additionally requires that excreta are safely contained, transported and treated (UNICEF & WHO, 2023). Improved facilities include flush or pour-flush toilets connected to piped sewer systems, septic tanks or pit latrines, ventilated improved pit latrines, pit latrines with slabs, and composting toilets.

Household sanitation encompasses more than the physical structure of a latrine. It includes the availability of water for cleansing, handwashing stations with soap, safe disposal of child faeces, maintenance of the facility, and hygienic behaviours of all household members (WHO, 2018). The World Health Organization's Guidelines on Sanitation and Health stress that sanitation is a system comprising hardware, such as pits, slabs and superstructures, and software, such as behaviours, norms, demand creation and operation and maintenance. Failure at any point of this service chain can re-expose households and communities to pathogens even after a latrine has been constructed (WHO, 2018).

In the Ethiopian context, household sanitation policy has evolved through successive health sector development programmes. The Health Extension Program placed

hygiene and environmental health among its sixteen service packages, and the model-household approach was introduced to graduate families through defined sanitation and hygiene criteria (Federal Ministry of Health, 2015). Community-Led Total Sanitation and Hygiene mobilizes communities to end open defecation through facilitated self-appraisal rather than external subsidies, while the One WASH National Program coordinates water supply, sanitation and hygiene investments across the ministries of water, health and education (Kar & Chambers, 2008; Ministry of Water, Irrigation and Energy, 2013). Within these frameworks, the household pit latrine remains the backbone of sanitation in both rural areas and secondary towns.

In urban settings the concept of household sanitation extends along the full sanitation service chain. Excreta from a household latrine must be contained in a hygienic pit, emptied when full, transported by manual or mechanical means, and treated or safely disposed of at a designated site. Weakness at any downstream link compromises upstream investments: pits that discharge into open drains, unregulated manual emptiers dumping sludge into rivers, or absent treatment facilities all convert nominally improved sanitation into environmental contamination. For secondary towns such as Mizan Aman, where onsite sanitation will remain dominant for the foreseeable future, utilization behaviour at the household level is the first and most decisive link of this chain.

2.2 Household Latrine Utilization

Latrine utilization refers to the actual and consistent use of a latrine facility for defecation and urination by all household members, as opposed to mere ownership or presence of the facility (Anteneh & Kumie, 2010). Utilization is a behavioural outcome that can be measured through self-reported use, observational indicators and triangulation of both. Commonly used observational indicators include signs of recent use such as wet footpaths leading to the latrine, fresh anal-cleansing materials inside the slab hole, absence of faeces on the floor, squatting plate or walls, absence of faeces in the compound and surrounding bush, and the general cleanliness and functional status of the facility (Yimam et al., 2014; Koyra et al., 2017).

Researchers have operationalized utilization in different ways. Some studies classify a household as utilizing its latrine when the facility shows clear evidence of regular use by most members during direct observation combined with a positive self-report (Anteneh & Kumie, 2010). Others grade utilization into categories such as full,

partial and non-utilization, depending on which members use the facility and under what circumstances (Koyra et al., 2017). For analytical simplicity, many community-based studies dichotomize utilization into utilized versus not utilized, an approach that will also be adopted in the present study and is described in detail in the operational definitions section of Chapter Four.

It is important to distinguish utilization from related concepts. Latrine coverage describes the proportion of households or population with access to a facility, regardless of use. Latrine ownership denotes possession of a facility by a household. Utilization, by contrast, captures behaviour over time and therefore requires maintenance, privacy, safety and social acceptability in addition to physical presence (UNICEF & WHO, 2023). Because administrative data in many districts report only coverage, the true public-health value of reported coverage figures can be overstated where utilization is not independently verified.

Measurement quality is central to utilization research. Self-reports alone tend to overstate use because latrine ownership carries social approval, while observation alone cannot reveal who within a household uses the facility or when. The most credible Ethiopian studies therefore triangulate: they ask structured questions about which members used the facility during the preceding days, inspect the facility for physical evidence of recent use, examine the compound and nearby bush for open defecation, and combine these signals into a summary judgement (Anteneh & Kumie, 2010; Yimam et al., 2014). This study adopts the same triangulation logic, as detailed in Sections F of the questionnaire and in the operational definitions.

2.3 Importance of Household Latrine Utilization

Consistent latrine utilization is one of the most cost-effective public-health interventions available to communities. Systematic reviews have shown that sanitation improvements reduce diarrhoeal morbidity by approximately one quarter to one third in low- and middle-income settings (Fewtrell et al., 2005; Wolf et al., 2014). Beyond diarrhoeal diseases, safe excreta disposal interrupts the transmission of intestinal helminths such as *Ascaris lumbricoides*, hookworm and *Trichuris trichiura*, reduces trachoma-transmitting eye seeking flies, and lowers the risk of typhoid and cholera outbreaks (Mara et al., 2010; Bartram & Cairncross, 2010). In children, reduced infection load contributes to improved nutritional status, growth and cognitive development.

The benefits of utilization extend beyond direct health outcomes. Safe sanitation protects water sources from faecal contamination, thereby reducing treatment costs for municipal supplies and the burden of waterborne disease at the community level (WHO, 2018). It enhances safety, privacy and dignity, particularly for women and girls who otherwise face harassment or snake and insect bites when defecating in the open after dark (UNICEF & WHO, 2023). School attendance among adolescent girls improves when home and school facilities are adequate, and community-level open-defecation elimination supports tourism, local markets and environmental quality (Kar & Chambers, 2008).

Economic analyses consistently demonstrate high returns on sanitation investment. Every dollar invested in sanitation is estimated to yield several dollars in avoided health-care costs, reduced premature mortality, and productivity gains, whereas poor sanitation imposes annual economic losses equivalent to several percent of gross domestic product in some developing countries (Bartram & Cairncross, 2010). These gains, however, are contingent on actual use: an unused latrine yields no protective effect, consumes scarce household resources, and may entrench complacency about open defecation (WHO, 2018).

Latrine utilization is also closely linked to educational outcomes and human-capital formation. Recurrent diarrhoeal episodes and intestinal worm infections cause school absenteeism, reduce concentration and learning capacity, and increase dropout risks among school-aged children. When households consistently use latrines and wash their hands, the cycle of reinfection within the family is broken, children attend school more regularly, and the benefits propagate into later productivity (Mara et al., 2010). Conversely, where latrines are constructed but left unused, health posts must continue diverting meagre resources toward treating preventable conditions rather than investing in development.

Furthermore, utilization carries important gender and equity dimensions. Women and girls bear particular costs when facilities are unusable: they may restrict daytime food and fluid intake, delay relief until darkness, and face risks of harassment or animal attacks when walking to distant open defecation sites (UNICEF & WHO, 2023). A private, safe and well-maintained household latrine restores dignity, enables menstrual hygiene management, and reduces the unpaid care burden that falls on women when family members fall ill. Consistent utilization is therefore not only a health intervention but also

a contribution to the gender-equality and reduced-inequalities commitments embedded in the Sustainable Development Goals.

2.4 Factors Affecting Household Latrine Utilization

The literature identifies a wide range of interrelated factors that determine whether constructed latrines are actually used. For analytical clarity, these factors can be grouped into five broad categories: socio-demographic characteristics, socio-economic conditions, knowledge and attitude, cultural and behavioural norms, and environmental or latrine-related attributes. The following sections review each category in turn.

2.4.1 Socio-Demographic Factors

Educational status is among the most consistently reported determinants of latrine utilization. Households headed by individuals with formal schooling are more likely to appreciate disease transmission pathways, to invest in facility maintenance and to abandon open-defecation habits. Studies in rural Amhara and southern Ethiopia found that latrine utilization increased significantly with the educational level of the household head or primary respondent (Anteneh & Kumie, 2010; Yimam et al., 2014; Koyra et al., 2017).

Age and sex of the household head or respondent also matter. Older respondents may adhere more strongly to traditional defecation practices, whereas younger heads may be more receptive to promoted behaviours; however, findings across studies are mixed regarding direction. Female decision-making influence within the household is frequently associated with better facility cleanliness and use, reflecting women's central role in domestic hygiene management (Jenkins & Scott, 2007). Larger family sizes create pressure on limited facilities, accelerate pit filling and may result in partial utilization, although some studies report the opposite association where larger families benefit from shared labour for maintenance (Koyra et al., 2017). Marital status, household composition, duration of residence and ethnicity have been examined less frequently but may operate through household resources and normative environments.

Duration of residence deserves particular attention in urban research. Recent migrants arriving from rural districts may carry strong open-defecation habits and unfamiliarity with compound sanitation, whereas long-established town residents have typically been exposed to repeated health-extension messages and municipal by-laws.

Household composition also shapes usage patterns: the presence of school-aged children has been associated with better utilization, possibly because children relay hygiene messages from school into the home, while households dominated by elderly members who grew up before latrine promotion campaigns may retain stronger field-defecation preferences. These considerations justify the inclusion of age, sex, education, family size, marital status and residence duration among the socio-demographic variables of this study.

2.4.2 Socio-Economic Factors

Income and wealth shape both the quality of latrine construction and the capacity to maintain it. Higher-income households can afford durable slabs, complete superstructures, doors and roofs, all of which enhance comfort, privacy and sustained use. Low-income households, by contrast, often construct temporary facilities that deteriorate rapidly during rainy seasons and fall out of use (Anteneh & Kumie, 2010). Occupation influences utilization indirectly through income stability and time availability; farming households engaged in field work away from home may practise open defecation near farmland despite owning a facility at home.

Access to water supply interacts closely with sanitation behaviour. Households without nearby water face practical difficulty in anal cleansing and cleaning the facility, which discourages use (WHO, 2018). House ownership is another relevant variable: tenants and landless residents may be reluctant to invest in improving facilities owned by landlords, resulting in neglected structures even in urban settings. Asset-based wealth indicators, such as livestock holdings, roofing material and durable goods ownership, have been used in Ethiopian studies as proxies for economic status and have shown significant associations with utilization outcomes (Central Statistical Agency & ICF, 2016).

Programme exposure is an additional socio-economic channel. Households reached by hygiene-promotion sessions, sanitation marketing or micro-finance schemes possess both the motivation and the means to keep facilities functional, and participation in such programmes has repeatedly differentiated users from non-users in Ethiopian field studies. In secondary towns, employment type also determines daytime presence at home: civil servants and traders who work near their residences find latrine use convenient, whereas daily labourers spending long days on distant construction sites may rely on whatever facilities exist at their workplace. Income stability further predicts the regularity

of consumable purchases such as soap, which supports the hand-washing behaviour that accompanies proper latrine use.

2.4.3 Knowledge- and Attitude-Related Factors

Knowledge about the health consequences of unsafe excreta disposal is a prerequisite for motivated latrine use. Respondents who can articulate how open defecation causes diarrhoea, intestinal parasites and other illnesses are more likely to use and maintain their facilities. Ethiopian studies report significantly higher utilization among households with good knowledge scores compared with those lacking such knowledge (Yimam et al., 2014; Koyra et al., 2017). Knowledge alone, however, is frequently insufficient, as documented by hygiene-promotion trials in which information did not automatically translate into practice.

Attitude towards latrine use shapes willingness to abandon habitual open defecation. Perceptions of latrines as smelly, dirty, frightening for children, or socially embarrassing discourage use, while perceived benefits such as comfort, safety, prestige and cleanliness encourage it (Jenkins & Scott, 2007). The theory of planned behaviour and related models emphasize attitudes, subjective norms and perceived behavioural control as proximal determinants of sanitation practices. Model households graduated through the Ethiopian health extension system typically display more favourable attitudes and correspondingly higher utilization than non-model households (Koyra et al., 2017), suggesting that attitude-centred programming can shift behaviour at scale.

The knowledge–behaviour gap is well documented: many respondents can list the diseases caused by open defecation yet continue the practice, indicating that knowledge interventions must be paired with elements that address emotional response, social expectation and convenience. Perceived susceptibility to and severity of faeco-oral disease, for example, determine whether abstract knowledge feels personally relevant. Similarly, perceived self-efficacy influences whether a household believes it can build, clean and maintain a latrine under its resource constraints. For these reasons, the present study measures both knowledge and attitude with multi-item scales rather than single questions, allowing their independent and combined contributions to utilization to be estimated.

2.4.4 Cultural and Behavioural Factors

Cultural norms surrounding defecation exert powerful and persistent effects. In many communities, open defecation is viewed as natural, healthy and socially preferable, particularly for morning visits to fields where people enjoy privacy, fresh air and social interaction. Long-standing habits formed before latrine construction are difficult to break, and adults may reserve latrines for guests, night-time use or illness while continuing to defecate outdoors (Kar & Chambers, 2008). Beliefs that sharing a latrine with in-laws is taboo, or that latrine use causes particular illnesses, have been documented in various settings and directly suppress utilization.

Behavioural preferences also include the type of anal-cleansing material, requirements for water availability, and fear of foul air in enclosed spaces. Children's faeces are sometimes regarded as harmless, leading to unsafe disposal practices even in latrine-owning households. Social prestige operates in both directions: a well-maintained latrine can signal modernity and status, encouraging use, whereas collective tolerance of visible open defecation normalizes the practice and undermines individual motivation (Jenkins & Scott, 2007). These culturally embedded factors justify their explicit measurement in the present study.

Gendered norms add a further layer of complexity. In many settings men control household investment decisions, including latrine construction and repair, while women manage daily use and cleaning; where women cannot voice facility deficiencies, latrines may remain unusable for months. Norms concerning modesty can also prevent adolescent girls from using facilities that lack secure doors, and expectations about hospitality may lead households to reserve clean latrines for visitors. Religious teachings generally support cleanliness, and religious leaders have proven effective allies in open-defecation campaigns; nevertheless, interpretations regarding acceptable places and times of relief vary and must be understood locally. Capturing such norms requires qualitative-sensitivity even within a quantitative instrument, which is why Section G of the questionnaire includes open-response items.

2.4.5 Environmental and Latrine-Related Factors

The physical characteristics of the facility itself strongly condition its use. Latrines that are structurally incomplete, lacking a roof, door or adequate superstructure, fail to provide privacy and protection from weather, and are consequently used intermittently or

abandoned. Dirty slabs, offensive odour, fly and mosquito breeding in pits, and fear of collapse deter household members, especially women and children (Anteneh & Kumie, 2010; Yimam et al., 2014). Distance between the dwelling and the facility matters: distant latrines are inconvenient at night and during rainy seasons, and long walking distances are associated with lower use, particularly by women concerned about personal security.

Facility age and maintenance history influence functionality. Newly constructed latrines may still coexist with residual open-defecation habits, while old latrines approach fill capacity and require emptying or reconstruction, which households may defer. Seasonality adds further complexity: high rainfall in Southwest Ethiopia can flood shallow pits, erode superstructures and render footpaths impassable, temporarily converting nominally utilizing households into open defecators. Sharing a facility between multiple households increases usage intensity, accelerates wear and complicates cleaning responsibilities, and shared facilities are commonly excluded from definitions of basic service coverage (UNICEF & WHO, 2023). Availability of water and soap near the latrine supports both use and handwashing, reinforcing the protective behaviour chain (WHO, 2018).

The neighbourhood environment matters as well. Households surrounded by visible open defecation receive a constant cue that undermines personal commitment, whereas streets where every compound maintains a clean latrine create positive peer pressure. Vegetation around latrines provides hiding places that facilitate sneaking to the bush, and has been associated with lower utilization in southern Ethiopian studies. Municipal factors, including solid-waste collection frequency, drainage quality and enforcement of sanitation by-laws, frame the broader environment within which household decisions are made. Because Mizan Aman's high rainfall intensifies several of these mechanisms, environmental and latrine-related variables receive detailed attention in the study design.

2.5 Review of Previous Studies

Table 1 summarizes selected previous studies on latrine utilization and its determinants conducted internationally and in Ethiopia. The reviewed investigations employed community-based cross-sectional designs and identified consistent associations between utilization and educational status, knowledge, attitude, economic variables and the physical condition of facilities.

Table 1: Summary of selected previous studies on household latrine utilization and associated factors

Author(s) / Year	Study Area / Setting	Design and Sample	Key Findings Relevant to the Present Study
Fewtrell et al. (2005)	Low- and middle-income countries (systematic review)	Systematic review and meta-analysis of WASH intervention trials	Sanitation improvements produced significant reductions in diarrhoeal morbidity; combined water, sanitation and hygiene interventions were advocated, underscoring that facilities must be used to confer protection.
Jenkins & Scott (2007)	Ghana (rural and urban households)	Cross-sectional household surveys on sanitation demand	Prior defecation practice, perceived convenience and social prestige strongly influenced latrine adoption and use; behavioural indicators predicted demand better than income alone.
Anteneh & Kumie (2010)	Hulet Ejju Enessie Woreda, East Gojjam, Amhara, Ethiopia	Community-based cross-sectional study among latrine-owning households	A considerable proportion of latrines showed no evidence of use; utilization was significantly associated with educational status, income and latrine condition, and proper use reduced diarrhoeal disease occurrence.
Yimam et al. (2014)	Denbia District, Northwest Ethiopia	Community-based cross-sectional study among 801 latrine-owning households	Utilization was satisfactory in about three fifths of households; maternal educational status, presence of a latrine door, hygienic condition and cleaning frequency were significantly associated with utilization.
Koyra et al. (2017)	Chencha District, southern Ethiopia	Community-based cross-sectional study among 420 latrine-owning households	Overall latrine use was reported by 60 percent of households but persistent utilization fell to about 31 percent; educational status of household heads, self-initiation, peer influence and time since construction were significant predictors.
EPHI & ICF (2021)	Nationally representative, Ethiopia	Ethiopia Mini Demographic and Health Survey 2019	National sanitation coverage remained uneven across regions and residence types, with large gaps between reported facility access and safely managed services.

Across these studies, common methodological features include community-based recruitment, combination of interviews with direct observation, and multivariable analysis of utilization as a binary outcome. Their consistency strengthens the plausibility of the factor categories adopted in the present study and provides benchmarks against which the Mizan Aman findings can later be interpreted.

2.6 Research Gap

Three principal gaps emerge from the reviewed literature. First, geographical gap: the overwhelming majority of Ethiopian latrine utilization studies have been undertaken

in rural districts of the Amhara, Oromia, southern and Tigray regions, while urban and peri-urban communities of Southwest Ethiopia remain largely unstudied. Second, contextual gap: existing studies focus on CLTSH-triggered villages and model households, whereas secondary towns experiencing rapid growth, tenancy, shared facilities and municipal service constraints present distinct utilization dynamics that rural-oriented instruments may not capture. Third, analytical gap: few studies have simultaneously examined cultural and environmental dimensions alongside socio-demographic and economic variables using observationally verified utilization outcomes.

No published study could be identified that quantifies the magnitude of household latrine utilization or its determinants in Mizan Aman Town. The present study addresses these gaps by applying a comprehensive instrument covering all five factor categories to a probability sample of latrine-owning households in the town, thereby generating locally actionable evidence.

Closing these gaps matters for practice as well as for scholarship. Without town-specific evidence, sanitation planners in Mizan Aman must borrow findings from dissimilar rural contexts, risking interventions that miss the barriers actually operating in urban compounds. By producing observationally verified utilization estimates and adjusted effect measures for each factor category, the study will supply exactly the kind of evidence that local programmes and the wider Southwest Ethiopian literature currently lack.

2.7 Underlying Theoretical Perspectives

Several behaviour-change theories inform the understanding of sanitation practices and the design of this study. The Health Belief Model posits that individuals adopt protective behaviours when they perceive themselves susceptible to a condition, judge the condition severe, believe the action beneficial, and see few barriers to performing it (Rosenstock, 1974). Applied to latrine use, households that perceive open defecation as genuinely threatening to child health, and latrine use as an effective and affordable response, are expected to utilize their facilities more consistently. The model explains why purely informative campaigns often fail when perceived barriers, such as foul smell or lack of privacy, remain unaddressed.

The Theory of Planned Behaviour extends this reasoning by emphasizing intentions as immediate predictors of behaviour, shaped by attitudes towards the behaviour, subjective norms and perceived behavioural control (Ajzen, 1991). In sanitation contexts, a household's intention to use its latrine depends not only on personal evaluation but also on whether respected community members expect such use and on whether the household feels able to maintain the practice. This perspective justifies measuring attitude scales and normative items alongside structural variables.

The Risks, Attitudes, Norms, Abilities and Self-regulation (RANAS) approach integrates these psychosocial factors into a practical framework for water, sanitation and hygiene programming, identifying which psychological factors steer a behaviour in a particular population and matching interventions accordingly (Mosler, 2012). Empirical applications of RANAS-style models in Southwest Ethiopia have demonstrated that attitude, descriptive norms and self-regulation significantly predict latrine use even after controlling for facility characteristics. Together, these theoretical perspectives support the study's decision to combine socio-structural variables with knowledge-, attitude-, norm- and facility-related measures in a single analytical model.

2.8 Conceptual Framework

The conceptual framework presented in Figure 1 was developed by the investigators through critical review of the literature summarized above, adapting elements from earlier sanitation behaviour frameworks (Jenkins & Scott, 2007; WHO, 2018). Five clusters of independent variables, namely socio-demographic, socio-economic, knowledge- and attitude-related, environmental and latrine-related, and cultural and behavioural factors, are hypothesized to influence the dependent variable, household latrine utilization. Potential confounders, including age, educational status, income and family size, will be controlled statistically through multivariable logistic regression.

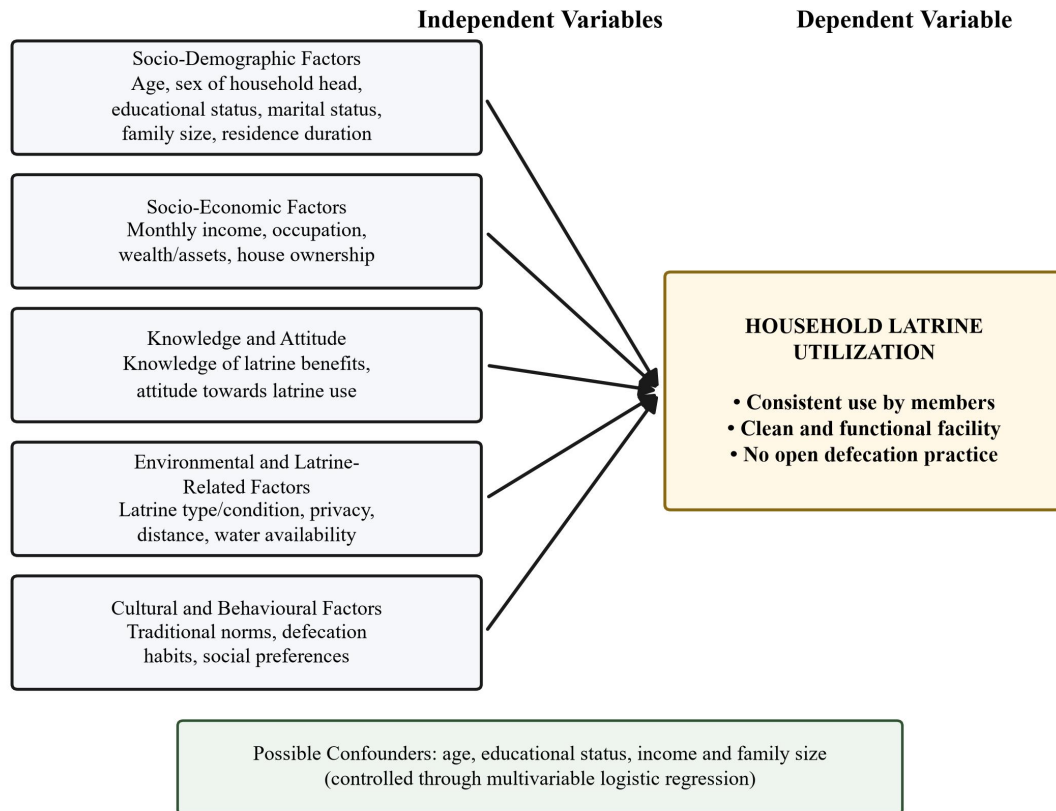


Figure 1: Conceptual framework of factors affecting household latrine utilization in Mizan Aman Town, 2026 (adapted from Jenkins & Scott, 2007 and WHO, 2018)

CHAPTER THREE: RESEARCH OBJECTIVES

3.1 General Objective

To assess the magnitude of household latrine utilization and identify the factors affecting it in Mizan Aman Town, Southwest Ethiopia.

3.2 Specific Objectives

Specifically, this study will:

1. Determine the magnitude of household latrine utilization in Mizan Aman Town.
2. Identify the socio-demographic factors associated with household latrine utilization in Mizan Aman Town.
3. Describe the socio-economic factors associated with household latrine utilization in Mizan Aman Town.
4. Assess the association between knowledge- and attitude-related factors and household latrine utilization in Mizan Aman Town.
5. Examine the environmental, latrine-related and cultural factors affecting household latrine utilization in Mizan Aman Town.

CHAPTER FOUR: METHODS AND MATERIALS

4.1 Study Area and Period

The study will be conducted in Mizan Aman Town, the administrative centre of Bench Sheko Zone, Southwest Ethiopia Peoples' Regional State. The town is located approximately 560 kilometres southwest of Addis Ababa and about 150 kilometres west of SNNPR's former zonal networks, connected mainly by the Jimma–Mizan and Bonga–Mizan all-weather roads. The town lies in a humid tropical agro-ecological zone characterized by high annual rainfall concentrated between March and October, which sustains coffee and spice farming in the surrounding districts.

Administratively, the town is organized into kebeles, the smallest formal government units, each led by a kebele administrator and supported by development agents and Health Extension Workers [VERIFY LOCALLY: exact number of urban and peri-urban kebeles, estimated total population, and total number of registered households from the Town Administration]. Health services in the town are provided by a primary hospital, health centre(s) and private clinics [VERIFY LOCALLY: current inventory of public and private health facilities from the Town Health Office]. Sanitation services rely predominantly on onsite facilities, principally simple and ventilated improved pit latrines, complemented by limited municipal solid waste collection [VERIFY LOCALLY: reported latrine coverage figures].

The study will be carried out from [Month, Year] to [Month, Year], covering proposal refinement, ethical clearance, community mobilization, data collection and analysis, as detailed in the work plan presented in Chapter Five.

4.2 Study Design

A community-based quantitative cross-sectional study design will be employed. The design is appropriate because it permits the measurement of the magnitude of household latrine utilization and the simultaneous assessment of multiple hypothesized explanatory factors at a single point in time. Self-reported information will be triangulated with direct observational checks of the latrine facility to strengthen the validity of the outcome measurement.

Alternative designs were considered and set aside. A cohort design, while stronger for causal inference, would require months of repeated household visits that exceed the academic calendar and budget available to the researchers; a qualitative design alone would not permit estimation of the magnitude of utilization or formal testing of associations; and analysis of secondary data is impossible because no town-level dataset on latrine utilization exists. The cross-sectional survey, executed with probability sampling and observationally verified outcomes, offers the most efficient balance of rigour, feasibility and cost for the objectives of this study.

4.3 Source and Study Population

Source population: All households residing in Mizan Aman Town during the study period.

Study population: All households in the selected kebeles that possess a latrine facility, whether privately owned or shared within the compound.

Statistical population: Adult household members aged 18 years or above who are knowledgeable about the household's sanitation practices, one of whom will be interviewed per sampled household.

4.4 Eligibility Criteria

4.4.1 Inclusion Criteria

Households will be included if they have resided in the study area for at least six months prior to the survey, possess a latrine facility, and have at least one adult member aged 18 years or above who is available and able to provide informed consent and respond to the interview.

4.4.2 Exclusion Criteria

Households will be excluded if the eligible respondents are critically ill or otherwise unable to communicate during the entire data collection period, if the household refuses to give informed consent after two consecutive visit attempts, or if the latrine was constructed less than one month before the survey, since such a facility would not yet reflect an established usage pattern.

4.5 Sample Size Determination

The required sample size will be determined using the single population proportion formula:

$$n = Z^2 \alpha/2 \times p (1 - p) \div d^2$$

where: n = minimum required sample size; $Z_{\alpha/2}$ = standard normal variate corresponding to a 95% level of confidence ($= 1.96$); p = assumed proportion of households utilizing latrines; and d = margin of error tolerated ($= 0.05$).

Since no published study quantifying household latrine utilization could be identified for Mizan Aman Town, the proportion will be taken as 50% ($p = 0.5$), which maximizes the required sample size and thereby guarantees adequate statistical power irrespective of the true local prevalence. Substituting the assumptions into the formula yields:

$$\text{Step 1: } n = (1.96)^2 \times 0.5 \times (1 - 0.5) \div (0.05)^2$$

$$\text{Step 2: } n = 3.8416 \times 0.25 \div 0.0025$$

$$\text{Step 3: } n = 0.9604 \div 0.0025$$

$$\text{Step 4: } n = 384.16 \approx 384 \text{ households}$$

$$\text{Step 5: Adjusting for an anticipated 10\% non-response rate: } n = 384 \times 1.10 = 422.4 \approx 423 \text{ households}$$

Therefore, a total of 423 households possessing latrine facilities will be included in the study.

4.6 Sampling Technique and Procedure

A multistage sampling technique will be employed to select the study households. In the first stage, all kebeles of Mizan Aman Town will be listed with support from the Town Administration, and [number] kebeles will be selected randomly by the lottery method [VERIFY LOCALLY: number of kebeles and selection decision approved with the advisor]. In the second stage, the total sample size will be allocated to each selected kebele proportionally to its number of latrine-owning households obtained from kebele household registration books [VERIFY LOCALLY]. In the third stage, the sampling interval k will be computed for each selected kebele as $k = N_i/n_i$, where N_i is the number of registered latrine-owning households in kebele i and n_i is the allocated sample. After

randomly selecting the first household by lottery, enumerators will proceed to every k^{th} household until the allocation is met. When a selected household is unavailable or declines participation after two visits, the nearest alternative household in the same direction will be visited. The overall sampling procedure is summarized in Figure 2.

To illustrate the allocation procedure, suppose that the town has K selected kebeles with registered latrine-owning households of $N_1, N_2, \dots, N_k$ and total $N = \sum N_i$. The share of the sample for each kebele will be computed as $n_i = (N_i \div N) \times 423$, rounded to the nearest whole number with adjustment so that the kebele-level allocations sum exactly to 423. Within each kebele the sampling interval will then be $k_i = N_i \div n_i$. All figures required for these calculations will be obtained from the current household registration books during community mobilization [VERIFY LOCALLY].

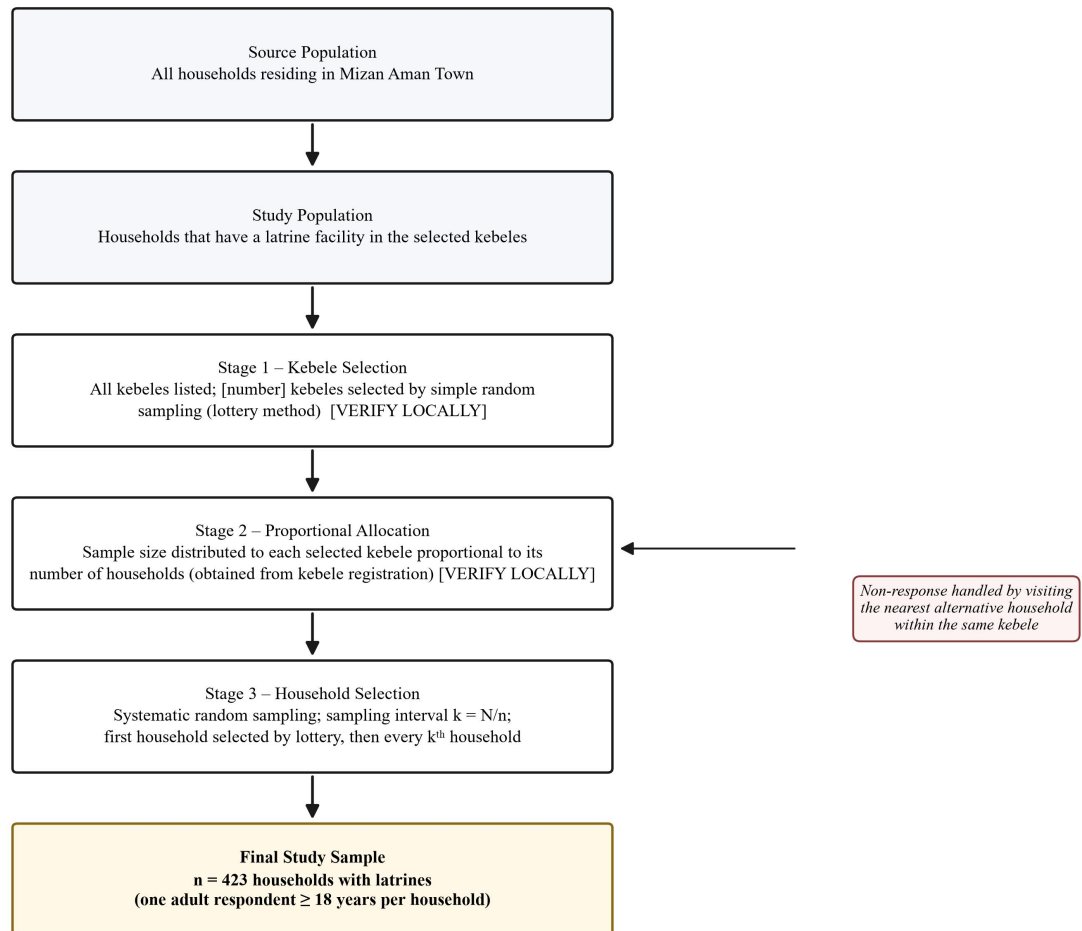


Figure 2: Schematic presentation of the sampling procedure for households in Mizan Aman Town, 2026

4.7 Study Variables

Dependent variable: Household latrine utilization (utilized / not utilized), measured by triangulating self-reported use with direct observational evidence.

Independent variables: Socio-demographic characteristics (age, sex, marital status, educational status, occupation, family size, religion, ethnicity and duration of residence of the respondent/household head); socio-economic variables (average monthly income, house ownership, asset ownership and water access); knowledge- and attitude-related variables (knowledge scores and attitude scores); cultural and behavioural variables (open-defecation habit persistence, normative preferences and child faeces disposal practice); and environmental and latrine-related variables (latrine type, age of the latrine, superstructure completeness, cleanliness, privacy, distance from dwelling, water availability near the facility and sharing status). These variables are summarized in Table 2.

Table 2: Summary of study variables, their measurements and operationalization

Category	Variables	Measurement Scale	Operationalization	Data Type
Dependent	Household latrine utilization	Nominal	Facility shows evidence of regular use by most members plus positive self-report	Binary
Socio-demographic	Age; sex; marital status; educational status; occupation; family size; religion; ethnicity; duration of residence	Years; categorical	Respondent report recorded in Section A of the questionnaire	Categorical / continuous
Socio-economic	Monthly income; house ownership; assets; water source and walking distance	ETB; categorical	Respondent report recorded in Section B	Categorical / continuous
Knowledge	Knowledge of latrine benefits and disease links	Score (correct answers)	$\geq$ mean score = good knowledge; $<$ mean = poor knowledge	Ordinal
Attitude	Perceived benefits, comfort and social value of latrine use	Five-point Likert scale	Sum score above median = positive attitude	Ordinal
Cultural / behavioural	Open-defecation habit persistence; normative preferences; child faeces disposal	Categorical	Respondent report recorded in Sections D and G	Categorical
Environmental / latrine-related	Latrine type; age of facility; superstructure completeness; cleanliness; privacy; distance; water	Categorical; metres	Direct observation checklist (Section F) verified by the enumerator	Categorical / continuous

	availability; sharing			
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4.8 Data Collection Tools and Techniques

Data will be collected using an interviewer-administered, structured and pretested questionnaire developed in English by the investigators after reviewing related literature (Annex I). The questionnaire contains seven sections: (A) socio-demographic characteristics; (B) socio-economic characteristics; (C) knowledge of respondents towards latrine use; (D) attitude of respondents towards latrine use; (E) environmental and latrine-related factors; (F) latrine utilization assessed through self-report and direct observation; and (G) cultural factors. An observation checklist (Annex IV) will be completed by the enumerator for every sampled household to verify the condition and usage evidence of the facility.

The English questionnaire will be translated into Amharic and, where necessary, into the predominant local language, and then back-translated into English by independent bilingual translators to ensure semantic equivalence. Six trained enumerators who have completed at least college-level education, together with two field supervisors and the principal investigators, will conduct the data collection. Enumerators will receive two days of intensive training covering the objective of the study, item-by-item review of the tool, interviewing techniques, observation procedures, research ethics and field logistics. The pretested tool will be administered face-to-face at respondents' homes at times convenient to them.

Two complementary data collection techniques will be used. The interview technique captures perceptions, knowledge, attitudes and self-reported practices that cannot be observed directly, while the observation technique provides objective verification of the facility's condition and usage evidence. Enumerators will record observational items only after personally inspecting the latrine with the respondent's permission, and will complete the summary classification item immediately before leaving the compound to minimize recall and courtesy bias. The combination of both techniques within one visit strengthens the validity of the dependent variable without imposing an excessive burden on respondents.

4.9 Pre-testing of the Instrument

Before the actual data collection, the questionnaire will be pre-tested on 21 households (approximately 5% of the calculated sample size) in one kebele of the town that will not be included in the main study. The pre-test will evaluate the clarity, length, translation adequacy and skip-pattern logic of the instrument, as well as the feasibility of the observational items. Based on the findings, ambiguous wording will be revised, unnecessary items removed and the time needed per interview estimated. The revised version will be reviewed and approved by the advisor before commencement of the main survey.

The pre-test will also serve logistical rehearsal functions. It will allow the investigators to test the household listing and sampling-interval procedure in field conditions, time each interview segment, observe how enumerators handle the observation items, and verify that translation conveys the intended meaning uniformly. A debriefing session on the final pre-test day will collect enumerators' impressions, and every proposed tool modification will be documented with its justification before the main survey begins.

4.10 Data Quality Assurance

Data quality will be assured throughout the study. During preparation, the tool will be expert-reviewed and pre-tested as described above. During collection, supervisors will accompany enumerators on the first day, thereafter conducting spot-checks and re-interviews on 10% of completed questionnaires; every completed questionnaire will be checked for completeness, consistency and legibility each evening, and feedback will be given to enumerators the following morning. Weekly progress meetings between the investigators, supervisors and enumerators will resolve emerging problems promptly.

During processing, questionnaires will be coded by trained data clerks and double-entered into EpiData Manager version 4.6 with built-in range and consistency validation rules. The two entries will be compared and discrepancies resolved against the paper originals. The cleaned dataset will be locked before analysis and stored with password protection accessible only to the investigators.

4.11 Data Processing and Analysis

Following completion of data collection, questionnaires will be checked visually for completeness, edited, coded and entered into EpiData version 4.6, and then exported to SPSS version 26 for cleaning and analysis. Descriptive statistics, including frequencies, percentages, means with standard deviations and medians where appropriate, will be used to summarize the socio-demographic characteristics of respondents and the magnitude of latrine utilization, and results will be presented using tables, charts and narrative text.

Bivariable binary logistic regression analyses will first be performed to examine the crude associations between each independent variable and latrine utilization; variables with a p-value of less than 0.25 in the bivariable analysis will be candidate variables for the multivariable model. Multicollinearity among candidate variables will be checked using the variance inflation factor, with values greater than 10 indicating collinearity problems. A multivariable binary logistic regression model will then be fitted using the enter method. The fitness of the final model will be tested using the Hosmer–Lemeshow goodness-of-fit statistic. Crude Odds Ratios (COR) and Adjusted Odds Ratios (AOR) together with their 95% Confidence Intervals (CI) will be computed, and variables with a p-value of less than 0.05 in the multivariable model will be declared statistically significant factors affecting household latrine utilization.

The analytical procedure will proceed through clearly ordered steps. First, the dataset will be explored for outliers, impossible values and missing patterns, and frequency distributions of all variables will be inspected. Second, the magnitude of latrine utilization will be computed as the proportion of households whose facilities met the combined self-report and observational criteria, together with its 95 percent confidence interval. Third, characteristics of respondents will be cross-tabulated against utilization status to provide an initial descriptive picture. Fourth, bivariable binary logistic regressions will estimate crude odds ratios, and variables satisfying the $p < 0.25$ screening rule will enter the multivariable model simultaneously. Fifth, model adequacy will be appraised using the Hosmer–Lemeshow test and classification accuracy, and adjusted odds ratios with their 95 percent confidence intervals will be reported for every retained variable. Model results will be presented in a single forest-style table so that readers can trace how crude associations change after adjustment for confounders.

4.12 Operational Definitions

Household: A person or group of persons who live together in the same dwelling unit, share meals and answer to the same household head.

Household head: The person recognized as such by household members and responsible for major economic and social decisions of the household.

Latrine facility: Any improved or unimproved excreta-disposal structure owned or shared by the household, including pit latrines with or without slabs and ventilated improved pit latrines.

Improved latrine: A flush or pour-flush toilet, ventilated improved pit latrine, pit latrine with slab, or composting toilet that is not shared with other households, consistent with JMP classification.

Functional latrine: A latrine that is structurally complete (slab, superstructure, roof and door present), not full or collapsed, and usable at the time of the visit.

Utilized latrine: A latrine showing observable evidence of recent regular use, namely absence of faeces in and around the facility, presence of anal-cleansing material or wet footpath signs, together with self-reported regular use by most household members during the preceding two weeks.

Non-utilized latrine: A latrine lacking the observational evidence described above or self-reported as not being used by most members.

Good knowledge: A respondent scoring at or above the mean of the knowledge items in Section C of the questionnaire.

Positive attitude: A respondent whose summed Likert attitude score in Section D is equal to or above the median score.

Model household: A household graduated through the health extension model-family package, as certified by the kebele Health Extension Worker.

Open defecation: Defecation in fields, bush, water bodies, open drains or any place other than a latrine, whether practised regularly or occasionally.

Safe child faeces disposal: Disposition of the faeces of a child under five years into a latrine or by burial, as reported by the respondent.

Shared latrine: A latrine facility used jointly by more than one household residing in separate dwelling units.

Rainy season: The period of heavy rainfall in the study area, approximately March to October [VERIFY LOCALLY with the Town Meteorological Service].

4.13 Ethical Considerations

Before commencement of the study, ethical clearance will be obtained from the Institutional Review Board (IRB) of Mizan-Tepi University [VERIFY LOCALLY: submission requirements and clearance reference number]. An official permission letter will then be sent through the University to the Bench Sheko Zone Health Department and the Mizan Aman Town Administration and Town Health Office to secure cooperation and access to the study kebeles.

During data collection, the purpose, procedures, potential risks and benefits of the study will be explained to each participant in the language they understand best, using the Participant Information Sheet (Annex II). Written informed consent (Annex III), including a thumbprint option for participants unable to read and write, will be obtained before any interview. Participation will be entirely voluntary, and respondents may refuse or withdraw at any time without any consequence to the services they receive. No invasive procedure will be performed. To protect confidentiality, no names or personal identifiers will be recorded on the questionnaire; instead, serial codes will be used, interviews will be conducted in private settings, and completed forms will be kept in a locked cabinet accessible only to the investigators. Electronic data will be password protected. Any hygiene advice requested by respondents will be provided briefly after the interview, and households found to have unsafe practices will be referred to the nearest Health Extension Worker.

Data protection will follow recognized good practice throughout the research life cycle: paper questionnaires will be transported in sealed envelopes and stored under lock; the electronic dataset will be anonymized at entry by retaining only serial codes; access will be restricted to named investigators; and all materials will be retained for a defined period after publication for audit purposes before secure destruction. No information enabling the identification of individual respondents will appear in any report, presentation or publication arising from the study.

4.14 Information Dissemination

The final report of the study will be submitted to the Department of Environmental Health, Mizan-Tepi University, and a copy will be deposited in the University library. Summarized findings and practical recommendations will be communicated to the Mizan Aman Town Administration, the Town Health Office and the Bench Sheko Zone Health Department through a feedback workshop. Where feasible, the findings will also be presented at scientific conferences and published in peer-reviewed journals to contribute to the wider sanitation literature.

4.15 Consistency Matrix

To ensure internal coherence among the objectives, variables, measurement instruments and analysis plans, the consistency matrix presented below maps each specific objective onto its key variables, the corresponding questionnaire section and the planned analytical method.

Specific Objective	Key Variable(s)	Questionnaire Section	Analysis Method
Determine the magnitude of household latrine utilization	Household latrine utilization (dependent)	Sections C, F and observation checklist	Descriptive statistics (frequency, percentage)
Identify socio-demographic factors associated with utilization	Age, sex, education, marital status, family size, residence duration	Section A	Bivariable and multivariable binary logistic regression
Describe socio-economic factors associated with utilization	Income, occupation, house ownership, assets, water access	Section B	Bivariable and multivariable binary logistic regression
Assess knowledge- and attitude-related factors	Knowledge score; attitude score	Sections C and D	Logistic regression; comparison of score categories
Examine environmental, latrine-related and cultural factors	Latrine type, condition, privacy, distance, sharing; norms and habits	Sections E, F and G plus observation checklist	Bivariable and multivariable binary logistic regression

4.16 Scope and Limitations of the Study

Scope: The study is delimited to households that already possess a latrine facility in Mizan Aman Town and to the factors affecting their utilization. Latrine coverage, construction quality standards across the whole town, faecal sludge management beyond the household compound and microbiological water-quality testing fall outside the scope of this proposal, although they constitute promising directions for subsequent research.

Limitations: First, the cross-sectional design measures exposure and outcome simultaneously and therefore cannot establish temporal sequences or causal relationships; associations identified should be interpreted as such. Second, self-reported utilization is subject to social-desirability bias, since latrine use is widely promoted as desirable behaviour; the study mitigates this limitation by triangulating self-reports with direct observational evidence. Third, seasonal variation may influence both facility condition and defecation practices, and findings collected in one season may not generalize to the other; the rainy-season timing will be documented and interpreted accordingly. Fourth, some sensitive cultural beliefs may be underreported in face-to-face interviews despite assurances of confidentiality. Finally, because the study excludes households without any latrine, its findings describe utilization behaviour among latrine owners only and should not be generalized to open-defecating households.

Despite these limitations, the combination of probability sampling, triangulated outcome measurement, enumerator training and multivariable adjustment provides a sound basis for drawing actionable conclusions for the town. The limitations will be explicitly acknowledged in all outputs of the study, and recommendations will be phrased with corresponding caution.

CHAPTER FIVE: WORK PLAN AND BUDGET

5.1 Work Plan

The proposed study will be completed within eight months from the approval of this proposal. The sequence of major activities and the months during which each will be undertaken are presented in Table 3 and visualized in the Gantt chart shown in Figure 3. The schedule allows adequate time for ethical clearance, instrument preparation, fieldwork and reporting while accommodating possible delays related to weather and respondent availability.

Table 3: Work plan of the study showing activities by month

No	Activity	M1	M2	M3	M4	M5	M6	M7	M8
1	Proposal refinement and advisor approval	✓							
2	Ethical clearance and permission letters		✓						
3	Tool adaptation, translation and pre-test		✓						
4	Recruitment and training of enumerators			✓					
5	Community mobilization			✓					
6	Data collection				✓				
7	Data entry, cleaning and verification					✓			
8	Data analysis and interpretation					✓			
9	Report writing and submission						✓	✓	
10	Dissemination of findings								✓

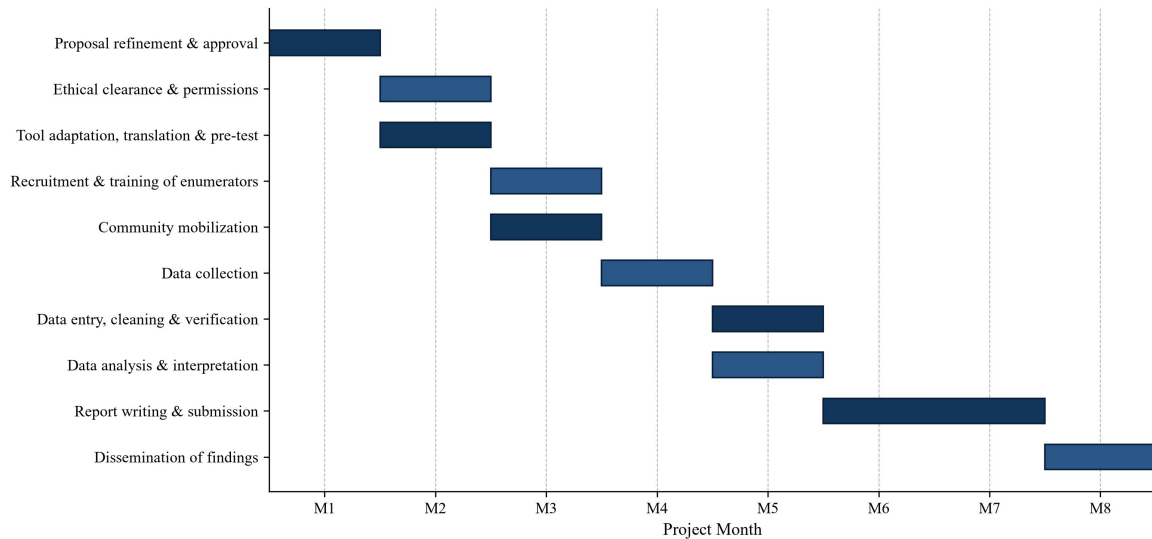


Figure 3: Gantt chart of the proposed study schedule

The schedule is organized into three phases. The preparatory phase (Months 1–2) covers proposal finalization, ethical clearance, tool translation and the pre-test; the fieldwork phase (Months 3–5) covers enumerator training, community mobilization through kebele leaders, data collection and supervised entry; and the reporting phase (Months 5–8) covers cleaning, analysis, write-up, advisor review cycles and dissemination. Float time has been built into Months 6 and 7 so that unforeseen delays in earlier activities do not jeopardize submission deadlines.

5.2 Budget Breakdown

The total budget required for the successful completion of the study is estimated at 69,520 Ethiopian Birr (ETB), including a ten percent contingency allowance for unforeseen costs such as price fluctuations and additional transport. The detailed breakdown is presented in Table 4. The researchers will cover any amount exceeding the estimate through personal contributions.

Table 4: Estimated budget breakdown of the study in Ethiopian Birr (ETB)

No	Budget Item	Description / Unit	Qty	Unit Cost (ETB)	Total (ETB)
1	Stationery	Paper reams, pens, markers, folders and envelopes	1 lot	3,000	3,000
2	Questionnaire printing	450 copies × 8 pages @ 2 ETB per page	3,600 pages	2	7,200
3	Training of data collectors	Venue, refreshments and training materials for 8 persons for 2 days	16 person-days	375	6,000

4	Enumerator allowance	6 enumerators $\times$ 10 days	60 person-days	400	24,000
5	Supervisor allowance	2 supervisors $\times$ 10 days	20 person-days	500	10,000
6	Transport	Vehicle rental and local travel for supervision	1 lot	8,000	8,000
7	Data clerks	Double data entry by 2 clerks $\times$ 5 days	10 person-days	300	3,000
8	Software and communication	Statistical software licence share, flash disks, airtime	1 lot	2,000	2,000
				Subtotal	63,200
				Contingency (10%)	6,320
				Grand Total	69,520

The budget has been prepared based on current market prices in the study area and may be revised with the advisor where necessary.

Cost efficiency has been pursued throughout the plan: questionnaires will be printed double-sided, enumerator routes will be clustered within kebeles to minimize travel time, and existing university statistical software will be used wherever licensing permits. The researchers will contribute their own laptops, and no honorarium is requested for the investigators. Should additional resources become available through departmental support, they will be allocated to enlarging the pre-test sample and strengthening supervision intensity.

CHAPTER SIX: REFERENCES

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ANNEX I: DATA COLLECTION QUESTIONNAIRE

Mizan-Tepi University – Department of Environmental Health

Questionnaire on Factors Affecting Household Latrine Utilization in Mizan Aman Town, Southwest Ethiopia

Instructions for enumerators: Read the consent statement aloud and obtain the respondent's consent before starting. Ask each question exactly as written, circle or tick the corresponding response code, and complete the observation items yourself without asking the respondent. Confidentiality must be strictly maintained.

Read-aloud introduction: “Good morning/afternoon! My name is _____ and I am working with researchers from Mizan-Tepi University. We are carrying out a study on how families in this town use their latrines. Your household has been randomly selected from this kebele. Everything you tell me will remain confidential and will be used only for research. Taking part is voluntary and you may skip any question or stop at any time. The interview takes about 30–40 minutes and includes a short look at your latrine facility. May I continue?”

Serial No.	Kebele	Household No.	Date	Enumerator Name
_____	[_____]	_____	___/___/___	[_____]

Section A: Socio-Demographic Characteristics

- A1.** Age of the respondent (in completed years): _____ years
- A2.** Sex of the respondent: 1) Male 2) Female
- A3.** Marital status: 1) Married 2) Single 3) Divorced/separated 4) Widowed
- A4.** Educational status: 1) Unable to read and write 2) Able to read and write only
3) Primary education (grades 1–8) 4) Secondary education (grades 9–12) 5)
College or university
- A5.** Occupation of the household head: 1) Farmer 2) Merchant 3) Civil servant 4)
Daily labourer 5) Housewife 6) Other (specify) _____
- A6.** Religion: 1) Orthodox 2) Muslim 3) Protestant 4) Catholic 5) Other
(specify) _____

- A7.** Ethnicity: [VERIFY LOCALLY: insert appropriate category list approved by the Town Administration] _____
- A8.** Total family size (number of members): _____ ; Number of children under five years: _____
- A9.** Duration of residence in Mizan Aman Town: _____ years
- A10.** Is the household certified as a model household? 1) Yes 2) No

Section B: Socio-Economic Characteristics

- B1.** Average monthly household income (ETB): 1) Less than 1,000 2) 1,000–2,999
3) 3,000–4,999 4) 5,000 or above
- B2.** Ownership of the dwelling: 1) Privately owned 2) Rented 3) Kebele/corporate housing 4) Other (specify) _____
- B3.** Main source of drinking water: 1) Piped connection in the yard 2) Public tap
3) Protected well or spring 4) Unprotected source 5) Other _____
- B4.** Total round-trip time to fetch water: 1) Less than 15 minutes 2) 15–30 minutes
3) More than 30 minutes
- B5.** Assets owned by the household (multiple responses possible): a) Radio b) Television c) Mobile telephone d) Livestock e) None
- B6.** Has any member participated in a hygiene or sanitation promotion session during the past year? 1) Yes 2) No; If yes, how many times? _____

Section C: Knowledge of Respondents Towards Latrine Use

Response codes: 1 = Yes; 2 = No; 3 = I do not know.

- C1.** Open defecation can cause diarrhoeal diseases. Response: _____
- C2.** Consistent latrine use protects against intestinal worms and other infections. Response: _____
- C3.** Faeces of small children should be disposed of safely in a latrine. Response: _____
- C4.** Hand washing with soap after using the latrine helps prevent illness. Response: _____
- C5.** A dirty latrine can transmit disease. Response: _____

C6. Flies can carry germs from faeces to food. Response: _____

C7. A latrine needs to be cleaned regularly to remain safe to use. Response: _____

Section D: Attitude of Respondents Towards Latrine Use

Please indicate the extent of your agreement with each of the following statements. Response codes: SA = strongly agree; A = agree; N = neutral; D = disagree; SD = strongly disagree.

No	Statement	SA	A	N	D	SD
D1	Using a latrine is more comfortable than defecating in the open.					
D2	Consistent latrine use protects my family's health.					
D3	Open defecation is an embarrassing and shameful practice.					
D4	I would encourage my neighbours to use their latrines.					
D5	I am willing to bear the effort of cleaning our latrine.					
D6	Latrine use is especially important for women's privacy and safety.					
D7	Spending money to maintain our latrine is worthwhile.					
D8	Children should be taught to use the latrine from an early age.					

Section E: Environmental and Latrine-Related Factors

E1. Type of latrine facility: 1) Simple pit without slab 2) Simple pit with slab 3) Ventilated improved pit 4) Flush/pour-flush toilet 5) Other (specify) _____

E2. Age of the latrine (time since construction): _____ months

E3. Who constructed the latrine? 1) The household itself 2) With programme or government support 3) Landlord 4) Other _____

E4. Completeness of the superstructure: roof present (1 = Yes, 2 = No); wall present (1 = Yes, 2 = No); door present (1 = Yes, 2 = No)

E5. Distance of the latrine from the dwelling: 1) Less than 6 metres 2) 6–10 metres 3) More than 10 metres

E6. Is the latrine shared with other households? 1) Yes 2) No; If yes, with how many households? _____

E7. Is water available within the household compound at present? 1) Yes 2) No

E8. Is anal-cleansing material available inside or beside the latrine? 1) Yes 2) No

- E9.** Who mostly cleans the latrine? 1) Mother/female head 2) Male head 3) Children 4) Shared by all 5) Nobody
- E10.** Frequency of latrine cleaning: 1) Daily 2) Only when it becomes dirty 3) Rarely or never
- E11.** Does the latrine currently need repair or maintenance? 1) Yes (specify) _____
2) No
- E12.** Status of the pit: 1) Not yet half full 2) Half to nearly full 3) Full/collapsed/abandoned

Section F: Latrine Utilization (Self-report and Observation)

- F1.** Do most members of this household use the latrine whenever they need to relieve themselves? 1) All members 2) Most members 3) Few members 4) Nobody
- F2.** Which members do not use the latrine? (multiple responses): a) Children under five b) Elderly members c) Adults who prefer the field d) None e) Others (specify) _____
- F3.** Reasons for non-use by any member (multiple responses): a) Fear of bad smell b) Distance from house c) Facility dirty d) No door/privacy e) Long-standing open-defecation habit f) Others _____
- F4.** In the past two weeks, did any household member defecate in the open environment? 1) Yes 2) No; If yes, under what circumstances? _____
- F5.** How are the faeces of the youngest child disposed of? 1) Into the latrine 2) Buried 3) Thrown into garbage 4) Open field 5) Not applicable
- F6.** OBSERVATION – Faeces visible on the floor, slab hole or walls of the facility?
1) Yes 2) No
- F7.** OBSERVATION – Faeces or fresh signs visible in the compound or surrounding bushes? 1) Yes 2) No
- F8.** OBSERVATION – Evidence of recent use (wet footpath, fresh anal-cleansing material)? 1) Yes 2) No
- F9.** OBSERVATION – Cleanliness of the slab: 1) Clean 2) Dirty 3) Very dirty

F10. ENUMERATOR SUMMARY – Overall classification: 1) Utilized 2) Partially utilized 3) Not utilized

Section G: Cultural Factors

G1. Before this latrine was constructed, was open defecation the main practice of the household? 1) Yes 2) No

G2. Do you ever feel that relieving oneself in the open is preferable? 1) Yes 2) No; If yes, why? a) Fresh air and comfort b) Social interaction c) Privacy in nature d) Habit e) No particular reason

G3. Are there cultural beliefs that prevent certain members (for example, in-laws) from sharing one latrine? 1) Yes (explain) _____ 2) No

G4. How does the community view families that own and properly use latrines? 1) Respected 2) Neutral 3) Mocked or criticized

G5. Are there seasons or occasions when open defecation becomes common in your area? 1) Yes (specify) _____ 2) No

G6. What suggestions do you have to improve latrine use in this community?

ANNEX II: PARTICIPANT INFORMATION SHEET

Title of the Study: Factors Affecting Household Latrine Utilization in Mizan Aman Town, Southwest Ethiopia.

Introduction: Good morning/afternoon! My name is _____. I am working with a team of researchers from the Department of Environmental Health, Mizan-Tepi University. We are conducting a study on household latrine utilization in this town, and we would like to invite your household to participate.

Purpose of the study: The purpose of this study is to assess how commonly latrines are actually used by households that own them and to identify the factors that encourage or discourage their use. The information collected will help the Town Health Office and partner organizations design better sanitation promotion programmes for this community.

Procedures: If you agree to participate, a trained enumerator will ask you questions lasting approximately 30 to 40 minutes about your household characteristics, your knowledge and views concerning latrine use, and your sanitation practices. The enumerator will also briefly observe the condition of the latrine facility. Your answers will not affect any service you receive.

Risks and discomforts: This study involves no invasive procedures. You may decline to answer any question you prefer not to answer, and the interview will be conducted in a private setting to protect your comfort.

Benefits: There is no direct payment for participation. However, the information you provide will contribute to improved sanitation programmes, and brief hygiene advice will be shared with you after the interview if you wish.

Confidentiality: No names or identifying details will be recorded. All information will be kept confidential, stored securely and used only for the purposes of this study in summarized form that cannot identify your household.

Rights: Participation is completely voluntary. You may refuse to participate or withdraw at any time during the interview without any penalty or loss of benefits.

Compensation: Participants will not receive any monetary payment for taking part. Light refreshments may be provided to respondents as a token of appreciation, and brief hygiene education materials will be shared with all participating households.

What happens after the study: The information collected from all participating households will be combined, analysed and summarized into a research report. A plain-language summary of the findings will be shared with the Town Administration and Town Health Office so that community members can also learn the overall results of the study.

Contact addresses: For questions about the study you may contact the principal investigators through the Department of Environmental Health, Mizan-Tepi University [VERIFY LOCALLY: insert telephone numbers and e-mail addresses]. You may also contact the Institutional Review Board of Mizan-Tepi University.

ANNEX III: INFORMED CONSENT FORM

Study title: Factors Affecting Household Latrine Utilization in Mizan Aman Town, Southwest Ethiopia.

Statement 1: “I confirm that the purpose, procedures, risks and benefits of this study have been explained to me in a language I understand.” **Initials:** _____

Statement 2: “I understand that my participation is entirely voluntary and that I may refuse to answer any question or withdraw at any time without any consequence.” **Initials:** _____

Statement 3: “I understand that the interview will take approximately 30 to 40 minutes and includes a brief observation of the latrine facility.” **Initials:** _____

Statement 4: “I understand that all information I provide will remain confidential and will be used only for research purposes in a form that cannot identify me or my household.” **Initials:** _____

Statement 5: “I voluntarily agree for my household to participate in this study.” **Initials:** _____

Role	Name	Signature / Thumbprint	Date
Participant	_____	_____	___/___/___
Enumerator	_____	_____	___/___/___

Note: For participants unable to read and write, the entire form shall be read aloud in the presence of an impartial witness, and consent shall be documented by thumbprint together with the witness's signature.

ANNEX IV: SUPPORTING MATERIALS

IV-A: Observation Checklist

To be completed by the enumerator through direct inspection of the latrine facility and compound.

No	Item Inspected	Standard Criterion	Result
1	Latrine structure standing	Walls, roof and slab intact	Y / N
2	Door present and functional	Closes adequately for privacy	Y / N
3	Slab cleanliness	No visible faecal soiling	Y / N
4	Faeces inside facility	Absent	Y / N
5	Faeces in compound or nearby bush	Absent within 50 metres of dwelling	Y / N
6	Evidence of recent use	Footpath signs or fresh cleansing material	Y / N
7	Anal-cleansing material available	Present within facility	Y / N
8	Water availability	Water source within compound or carried to facility	Y / N
9	Pit condition	Not full, collapsed or abandoned	Y / N
10	Fly breeding	Minimal or absent	Y / N

IV-B: Permission Letter Template

[Mizan-Tepi University letterhead]

To: Mizan Aman Town Administration / Town Health Office
Subject: Request for Permission and Cooperation

This is to certify that the students whose names appear on the cover page of this proposal are bonafide students of the Department of Environmental Health, Mizan-Tepi University, and are conducting a research study titled “Factors Affecting Household Latrine Utilization in Mizan Aman Town, Southwest Ethiopia.” The University kindly requests your office to grant the team permission and the necessary cooperation. [VERIFY LOCALLY: official reference number, signature and stamp.]

IV-C: Ethical Clearance Certificate

[Insert scanned copy of the Institutional Review Board ethical clearance certificate issued by Mizan-Tepi University here once obtained. VERIFY LOCALLY after submission of the proposal.]

IV-D: Data Quality Assurance Checklist

Stage	Quality Control Activity	Responsible Body
Pre-field	Expert review and translation check of the tool	Investigators and advisor
Pre-field	Pre-test on 21 households and tool revision	Investigators
Field	Two-day enumerator training with practical exercise	Investigators, supervisors
Field	Spot-checks and re-interview of 10% of respondents	Supervisors
Field	Daily completeness and consistency checking	Supervisors
Post-field	Double data entry with validation rules	Data clerks
Post-field	Range and consistency verification of the dataset	Investigators

IV-E: Enumerator Training Agenda

Session	Content	Duration
1	Welcome; overview of the study objectives and significance	1 hour
2	Research ethics: consent, confidentiality, voluntariness and respondent dignity	2 hours
3	Item-by-item review of the questionnaire and observation checklist	4 hours
4	Interviewing techniques, probing without leading, and handling refusals	2 hours
5	Sampling procedure practice: household listing, interval selection and replacement rules	2 hours
6	Field practicum with role-play and mock interviews	3 hours
7	Daily supervision plan, data flow and quality checks	1 hour